



Graphs in Machine Learning

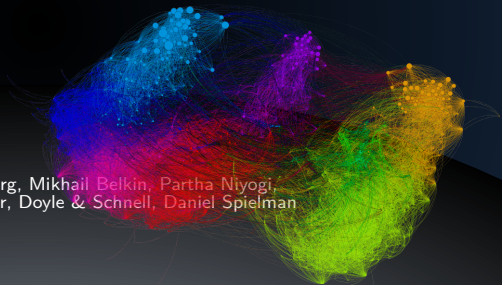
Manifold Learning

Laplacian Eigenmaps

Michal Valko

Isara Labs

Partially based on material by: Ulrike von Luxburg, Mikhail Belkin, Partha Niyogi, Olivier Chapelle, Bernhard Schölkopf, Gary Miller, Doyle & Schnell, Daniel Spielman



Background: Manifold Learning

problem: definition reduction/manifold learning

Given $\{\mathbf{x}_i\}_{i=1}^N$ from $\mathbb{R}^d$ find $\{\mathbf{y}_i\}_{i=1}^N$ in $\mathbb{R}^m$, where $m \ll d$.

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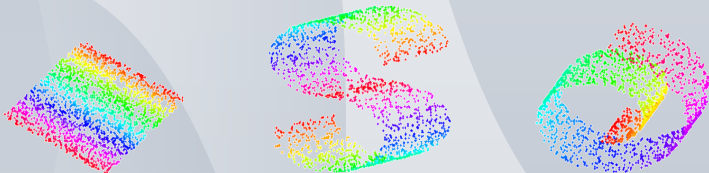
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 - **nonlinear often preserve only local distances**

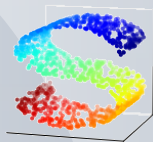
Manifold Learning: Linear vs. Non-linear



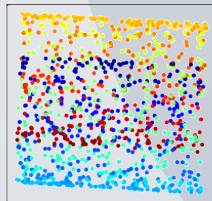
Source: Belkin & Niyogi (2003)

Manifold Learning: Linear vs. Non-linear (Alternative View)

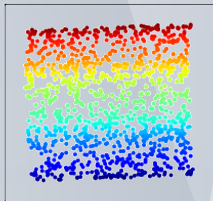
What do we know about linear vs. nonlinear methods?



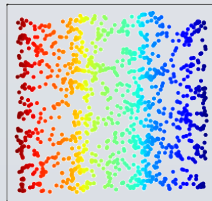
PCA projection



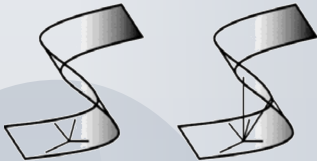
LLE projection



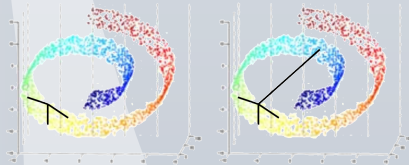
IsoMap projection



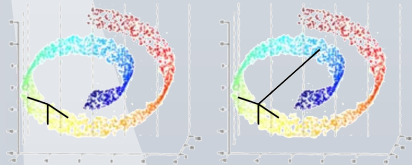
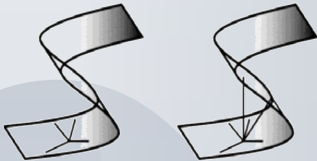
Manifold Learning: Preserving (just) local distances



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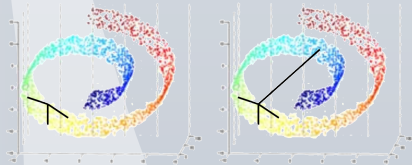
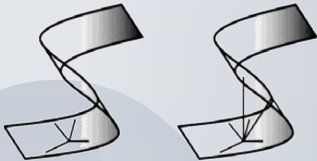
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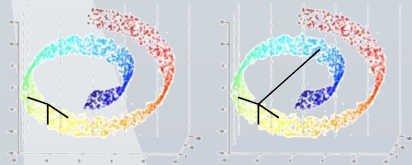
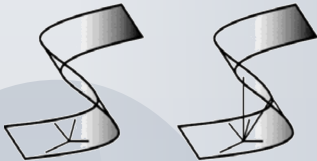


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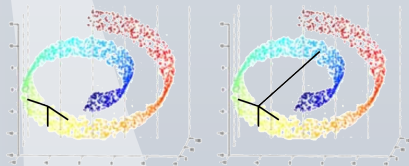
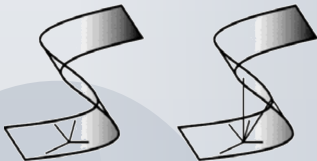
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Looks familiar?

Manifold Learning: Laplacian Eigenmaps

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Note₂: $\mathbf{f}_1$ is useless

http://web.cse.ohio-state.edu/~mbelkin/papers/LEM_NC_03.pdf

Manifold Learning: Laplacian Eigenmaps to 1D

Laplacian Eigenmaps 1D objective

$$\min_{\mathbf{f}} \mathbf{f}^T \mathbf{L} \mathbf{f} \quad \text{s.t.} \quad f_i \in \mathbb{R}, \quad \mathbf{f}^T \mathbf{D} \mathbf{1} = 0, \quad \mathbf{f}^T \mathbf{D} \mathbf{f} = 1$$

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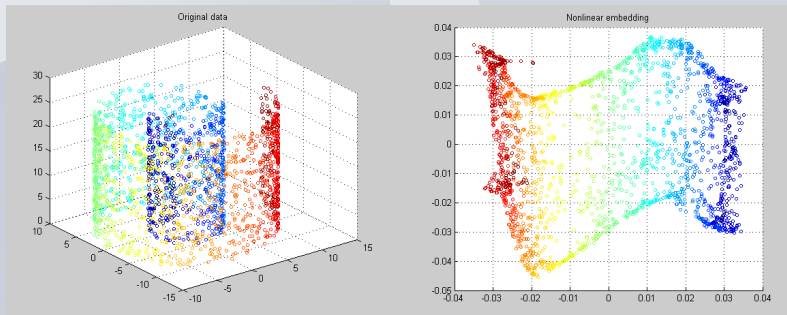
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What is the solution?

Manifold Learning: Example



Source: Belkin & Niyogi (2003); MATLAB implementation: <http://www.mathworks.com/matlabcentral/fileexchange/36141-laplacian-eigenmap---diffusion-map---manifold-learning>

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<https://misovalko.github.io/mva-ml-graphs.html>